Software Engineer

Intro

Functions vs Procedures

Functions

The functions are routines which perform some work or specific algorithm to solve the problem and return one of the following values:

- **Value (Number, Alphanumeric, Etc.)**
- **Function (Object Function)**
- **Array**
- **Set of Values**

Procedures

The procedures are routines which perform a work or algorithm meanwhile other functions do not return any value, receive parameters, each one of them is a different type of data, for example:

- **Int**
- **Double**
- **Float**
- **Date**
- **Char**

Procedures

The routine is a series of commands where it performs an algorithm or problem, and it tries to solve such problem.

This routine does not return any value.

It receives parameters, and some of them are optional, others are mandatory.

Classes

Encapsulation

The encapsulation is for containing into one entity everything which it can do such entity, for example:

- **Class**
  - **Car**
- **Property**
  - **Color**
  - **Type of Insurance**
- **Method**
  - **Turn On**
  - **Stop**
  - **Park**

Constructor

The constructor is a intrinsic method of work which perform when it creates the instance of object.

At the moment that creates an instance of class, it calls to this method by default.

This used commonly to create the instance of the object.

The constructor can be overwritten in different ways so that it can give different instances of the same object, and from there, it can be called in different ways for creating different instances of the same class.

Constructor

It means that it can create different similar objects of the same class.

Destructor

The destructor is an intrinsic method that calls when it destroys the instance of an object.

It is useful to free an space of memory and of this manner which the program do not full of trash.

In this way, the program can work in a light way without not so much load and this is the way that can continue till finishes the program.

Destructor

In the contrast, it can full much memory or make use of memory without using it, and there so, it can finish the program much before.

And this does not happen very much when it works with volumes of data or tasks very big.

The program can finalize in different ways and better performance.

This method is important. The program could delay in many occasions without it.

Method

Method is an action of the class. It can be called after it creates an instance of the object.

So there each method can be called to solve some particular problem, and you can create as many methods as possible.

Each one of them will perform a different task and it will solve many problems during the execution of the program.

Method

Such beginning or ending of program, is very important its task.

Because of that, it is used such many times during the running of the program.

If inside this method, it is used many resources of the system, we should consider that we have something to free those resources because they are blocks or chunks or memory.

Visibility

Visibility

The visibility is a feature that each property or method of the class has, for example:

- **Private**
  - **It means when it can not be accessed from outside. Only it can be called from the class.**
- **Protected**
  - **It means when it can not be accessed from outside. Only it can be called from the same instance of the object.**

Visibility

- **Public**
  - **It means when it can be accessed from outside. After the creation of the instance of the object.**

Overwrite

It means when it repeats the same method but with more parameters.

In other words, it has the same signature.

It can be overwritten such many times as you wish, however we should be careful with it.

Because if we repeat it many times, it could create confusions at the moment that another developer tries to continue it.

Override

It means when it repeats the same method but it has the same parameters.

In other words, and a bit more technical it has the same signature but with different <type> of parameters.

However, you need to pay attention because it can be confusing for other developers who need to continue or modify the method implementation.

Inheritance

It means when data inherits the same method and/or properties from the parent class.

This is used frequently to reuse the same functionalities from the classes which are levels above.

You do not need to write the code again.

Because of this, the program codification speeds up.

Interface

It is used for creating an interface that the class needs to respect.

The class must contain the same methods and/or properties that the interface contains.

Then, the children classes will have the same inheritance.

Polymorphism

It is used to overwrite the parent class.

In this way, the parent class is an abstract class and its children classes are changing slightly to different shapes, using different properties and methods.

So, every child class inheritance will be different from the original parent class.

Meaning of Data

Meaning of Data

Data has its own meaning inside a Company/Business/Small Business.

Meaning of data is crucial inside a Company.

Because decisions making is mush easier and it gives a better direction to the Company's businesses.

From this start, people who make decisions, know which resources need to hire and know which tasks need to carry on, to choose the most convenient place for the Company.

Meaning of Data

People, who can make decisions, use this data to go in the more suitable direction and lead his/her Company to a safe place.

Or even though, these people make Company continue doing business, as many years as possible.

Unfortunately, Some people can make it and others cannot.

People, who make decisions, are responsible of this situation.

Because of this, people who are on this business level, may perform this work type and underneath this data, people are involved.

Tables

Tables are a set of records which are stored inside a database.

Such tables may relate each other by its fields, or even though, by this database tables.

Tables is a place where Company's records are stored.

Each Table may incorporate new fields, changing its original shape.

Programming Languages Comparison

Web Development

Web Development is used to show more easily what it is trying to sell to the Client.

Nowadays, entrepreneurs use many different strategies to commercialize the product much easier.

This type of development is applied in different working fields, due to its portability among different Operating Systems and mobile devices.

Desktop Development

Desktop Development is used where Operating System is nearly obsolete.

Unfortunately, this type of development is being slowly discarded because Software Development is currently being oriented to Web Development or Mobile Development.

Process Development

Process Development is used to automate the tasks so that these ones might finish more efficiently.

Then, it is not needed to repeat the tasks again.

Database Development

Database Development is used to manipulate data in a efficient and more straightforward way.

Then, there are different developments to do the same routine or task.

Problem Solving

Analysis

Analysis is about the understanding of the problem and how to divided up into parts so that it can be much easier to solve it.

The ones, who are going to develop and solve the problem, might understand how to code it in the best possible way.

Although, at times, software development implies to start coding in the worst way.

Because client needs to see the application running as quick as possible.

Documentation

Documentation is about documenting code and analyzing the software which is going to be develop.

Code can be modified and understood by the rest of work-mates.

Where each work-mate may have a better overview about changes that must be done.

Documentation

Documentation is useful because it may explain how to continue coding without any important delivery delays.

However, it also may be used to know which software development alternatives actually exist and review if the development is going to another direction which was not planned at the beginning.

Architecture Diagramming

Architecture Diagramming is about showing how the software works from start to end.

The main purpose is understanding how to create and code the software.

Then, every member team might perform the software in a more productive way.

Architecture Diagramming

Performing this Diagramming, every member team can visualize how all pieces are going to be joined at the end of the project.

Sometimes, this kind of work is underestimated because it implies to design the Software Architecture and show the right way to build up a Software.

Coding

Coding is useful to command computer what it should do and how it should be done in the most efficient way.

Every short software delivery must be even shorter along the project.

Every member team need to work with the rest as a team.

If not, it is hard to accomplish every delivery.

First Software Delivery

When development team performs the first delivery.

Then, they can go ahead with the following pending deliveries.

This first delivery may get complicated because coordinating, professionals who are involved, could be a difficult task to achieve.

Client Evaluation

When client needs to evaluate how the first delivery was done.

Then, client observes what it needs to improve or remove from that delivery.

At times, client does not know what he really needs.

This is not something to be considered as important.

In fact, client does not know very clearly what he is asking for.

Because of this, client asks a group of experts who can develop the software.

First Delivery Feedback

The feedback of the first delivery should be done as much quick as possible.

Because unfortunately there is no place to make important improvements around.

Company needs the job done as soon as possible.

Because of this, some professionals want to work inside or outside a Company.

Feedback as many times as possible to reach definitive delivery

Feedback or deliveries are usually too many because it requires corrections in each delivery till reaching the end of final product release.

However, the final product release ends being something that it is not so clean as it is expected.

Client needs the product to start commercializing its own products.

Unfortunately, client does not care about how many lines of coding, is behind such product.

Data Visualization

Data Visualization

This is useful to see how data behaves inside a company, business, or entrepreneurship.

Also, it is useful to analyze the problem, which should be solved, from different angles.

This is crucial to make decisions because it becomes easier to sell the product.

Data Visualization

Software optimization and problem solving is also much easier.

In this way, the product can be observed from different projections in a graphical mode.

On the other hand, the period of software development gets shorter and the tasks to be accomplished are not always finished.

English Language

Virtual Meetings

This meetings can be difficult to manage if there is no experience about it.

In fact, each member's meeting should be granted to talk or not, depending on who is going to start talking.

This new paradigm we are getting in, little by little.

Everyone should be get used to it although some professionals do not really like it much.

Virtual Meetings

Old human generations continue accepting old fashioned meetings as usual.

However, everyone may choose what to do with it.

Technology keep going forward more and even more and each of us gets into this technology world.

By the way, we are already inside it.

Although, some professionals do not want to accept it.

Old Fashioned Meetings

These meetings were oriented to professionals who want to communicate personally.

Actually, there are still some professionals who continue working using these kind of meetings.

Old fashioned meetings keep continue being even more healthy for many professionals.

However, professionals might choose other meetings kind where they feel more comfortable.

Old Fashioned Meetings

These meetings are useful to talk in a more communicative way about a topic.

Each member's meeting has the right to express as he/she wants to.

However, there is someone who moderates the meeting.

Because, there could not be any meeting, without it.

There may exist some disagreements but it is what is supposed to happen.

Without such disagreements, meetings could not reach any agreement.